

How Long Is a Thought? Separating Observer-Measured Duration from Temporal Use and Reconstruction

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Abstract

The question “How long is a thought?” contains several distinct measurement objects. An observer can assign a bounded cognitive episode an elapsed duration, but that duration and a later report do not, without additional assumptions or evidence, identify whether metric temporal information was causally used while the episode unfolded. This theoretical article distinguishes observer-measured duration (T_O), latent online temporal use (U_T), a post-episode temporal basis/access/reconstruction category (A_{post}), the first post-query report ($\hat{T}_Q$), the duration of a deliberately induced reflection episode (T_R), and its later report ($\hat{T}_R$). The key problem is observational equivalence: the same objective/report pair can arise from online monitoring, online-but-unused temporal information, post-query reconstruction from event structure or other cues, or mixed routes. A minimal protocol preserves one temporally naïve bounded episode, obtains a minimally elaborated post-query duration judgment, times a deliberate act of thinking about the original episode’s duration, and then obtains a quick estimate of that reflection episode. It intentionally does not collect a second estimate of the original episode. The recursive probe demonstrates that objective-duration/report non-identification recurs at the next level; it is not a unique mechanistic discriminator. Existing work on

retrospective/prospective timing, quantified introspection, temporal metacognition, and memory-based duration judgment strongly precedents the components. The residual contribution is the integrated identification structure and staged measurement sequence.

Keywords: time perception, duration judgment, retrospective timing, prospective timing, temporal metacognition, introspection

1. Introduction: How Long Is a Thought?

A person begins to think and, some time later, the thought is over. From outside the person, the interval can in principle be timed. If a task cue marks the beginning and a key press or verbal signal marks completion, an observer can assign the episode an elapsed duration. That measurement is straightforward in one sense: it is a property of the bounded episode as operationally defined by the experiment. The harder question is what, if anything, that elapsed duration tells us about temporal information represented, accessible, or causally used by the person while the episode was occurring.

The distinction matters because the same phrase - “how long did that thought take?” - can refer to different scientific objects. It can mean the elapsed interval between experimentally defined onset and offset. It can mean whether metric elapsed-time information entered the ongoing cognitive process. It can mean what temporal basis can be accessed or constructed only after a query is posed. Or it can mean the numerical or verbal estimate eventually reported. These objects can covary without being identical.

Timing research already provides a mature distinction between prospective and retrospective duration judgments. In prospective timing, participants know in advance that duration is relevant; in retrospective timing, the duration query is unexpected until after the target interval. Reviews and meta-analyses have shown that these paradigms recruit attention and memory differently and can yield different judgments (Block & Zakay, 1997; Grondin, 2010). More recent work preserves the same practical distinction: a retrospective judgment is collected before participants know that timing will be queried, whereas subsequent trials become prospective or temporally informed once that fact is known (Nicolai et al., 2024).

That literature motivates, but does not by itself settle, a narrower identification question. If a participant produces a retrospective estimate after a naïve cognitive episode, the estimate demonstrates that the participant can generate a temporal judgment after the query. It does not,

without additional evidence, establish that metric temporal information participated causally in the original thought, nor does it establish that the basis of the report existed in the same form before the query. The reverse inference is also unsafe: failure to report a duration does not prove that no temporal information was present or used. Reporting depends on memory, interpretation of the question, response mapping, confidence, and other post-episode operations.

Prospective timing does not remove the problem. Informing a participant that time matters may make temporal monitoring part of the cognitive episode. That can be scientifically useful, but it changes the target. A prospective trial can tell us how cognition behaves when duration is task-relevant; it cannot automatically recover the temporal organization of the same episode before timing became relevant.

This article develops that distinction as an identifiability and measurement problem rather than a metaphysical claim. It does not argue that thoughts have no duration, that consciousness lies outside physical time, or that people cannot estimate the duration of their own cognition. The claim is narrower: observer-measured duration and a later duration report do not, without additional assumptions or independently validated measurements, identify whether metric temporal information was causally used during the original naïve episode.

The constructive part of the article is a simple staged protocol. A participant first completes a bounded cognitive episode without advance notice that elapsed duration is the target. After completion, the participant gives a quick post-query duration estimate. The participant is then instructed to think deliberately about how long the original episode took; that duration-focused reflection is itself externally timed. Finally, the participant gives a quick estimate of the reflection episode's duration. No second estimate of the original episode is collected. The recursive step is not intended as a paradox or a mechanistic discriminator. It makes visible that the operation introduced to reason about one timed episode is itself another timed episode whose later report faces the same identification problem.

The resulting framework is intentionally modest but theoretically consequential. It treats ordinary objective-duration-plus-report data as observationally compatible with more than one latent temporal organization. The proposed sequence exposes where new cognitive operations enter the measurement chain and what remains latent. The goal is not to replace established theories of time perception but to clarify what this family of measurements can and cannot identify about a bounded cognitive episode.

2. Existing Timing Paradigms and the Identification Gap

Prospective and retrospective timing are the nearest established starting point. Block and Zakay's (1997) meta-analysis found systematic differences between judgments made when participants expected to judge duration and judgments made after an interval whose duration had not been task-relevant. The broad theoretical lesson is familiar: prospective judgments are especially sensitive to attention allocated to time, whereas retrospective judgments can depend heavily on memory for events, contextual changes, and information accumulated during the interval.

Grondin (2010) reviews a larger timing literature in which explicit duration judgments can be supported by several mechanisms rather than by a single, universally accessed internal clock. The distinction is particularly important for naïveté. Nicolai et al. (2024) used a working-memory task in which the very first n-back sequence was temporally naïve and its subsequent duration judgment was retrospective; after that first query, remaining trials were treated as prospective because participants knew timing judgments would follow. The study therefore provides a concrete precedent for the naïve-to-informed transition that matters here. Their design was not intended to isolate the present identification problem, and first-versus-later differences remain entangled with order and task familiarity, but it shows why repeated duration questions cannot all be treated as equivalent surprise-retrospective observations.

A second relevant literature uses quantified introspection to compare objective processing times with subjective estimates. Corallo et al. (2008) showed that introspective response-time estimates can track objective response time closely in a single-task context, yet a processing delay introduced by a dual-task bottleneck can be absent from the subjective estimate. Marti et al. (2010) extended this approach and found that introspective estimates could preserve some components of processing time while largely missing the period in which a second target waited during the psychological refractory period. These studies are important for two reasons. First, they reject an indiscriminate skepticism about introspective timing: subjective temporal reports can contain real chronometric information. Second, they show why report accuracy cannot be equated with complete access to the temporal structure of the underlying process.

Recent work explicitly treats timing reports as metacognitive measurements. Pavallier et al. (2025) asked whether introspective reaction-time judgments reflect direct access to processing duration or inference from other cues; trial-by-trial introspective and objective reaction times were strongly related, yet judgments were also systematically biased by nontemporal conditions.

Cropper et al. (2024) likewise collected a first-order judgment of perceived interval duration together with a second-order metacognitive judgment of perceived time. These studies establish close precedents for temporal metacognition and for separating first-order temporal behavior from second-order temporal judgment. They do not, however, make a later report a transparent copy of the temporal variables causally used during a prior naïve cognitive episode.

Philosophical work on the passage and perception of time also provides a useful foil. Phillips (2012, 2013) argues that concurrent non-perceptual conscious mental activity can contribute to perceived duration and to relative temporal judgments. That possibility is compatible with the present framework. Ongoing mental activity might provide temporal information, event structure, or cues that are represented online, causally used online, or accessed only after a query. The identification point is simply that the presence of changing conscious content does not, by itself, establish that metric elapsed duration was a variable causally used by the naïve thought.

Experience-sampling methods make a complementary methodological point. Descriptive Experience Sampling attempts to minimize presupposition and asks participants to capture experience at unpredictable moments (Hurlburt & Akhter, 2006). Sutton (2011) notes a tradeoff: snapshot sampling can protect an experience from some forms of anticipatory measurement while also discarding temporally extended dynamics. Historical work on spontaneous thought segments further shows the difficulty of defining natural thought boundaries. Klinger (2013), reviewing earlier thought-sampling studies, notes both retrospective estimates of thought-segment duration and the lack of a fully satisfactory method for determining when naturally occurring thought segments begin and end. Those precedents make two constraints unavoidable here: the episode boundary must be operational rather than assumed to reveal a natural “atom” of thought, and the temporal query itself must be treated as an intervention on later cognition.

Studies of retrospective duration during internal and naturalistic tasks also show that later judgments can depend on memory without answering the online-use question. Vitulli and Shepard (1996), for example, measured retrospective estimates after arithmetic and recall tasks and found effects of task and delay. More recently, Yue et al. (2026) found that retrospective judgments of naturalistic event duration depended on remembered event structure, especially recalled subevent boundaries for whole events. Such findings establish memory-based routes to later duration judgments. They do not establish which metric temporal variables, if any, were causally recruited while the original cognitive activity unfolded.

The residual gap is therefore narrow. Existing research already establishes prospective-versus-retrospective differences, memory- and attention-sensitive duration judgments, quantified introspective timing, temporal metacognition, and measurement reactivity. What remains open is the integrated identification structure: how observer-defined duration, latent online temporal use, post-query access or reconstruction, and later reports should be related when the target is a temporally naïve bounded cognitive episode. The protocol developed below does not identify every latent quantity. Its purpose is to type them explicitly and stage the observable operations so that later inference does not collapse them by default.

3. Measurement Objects and Latent Structure

The central proposal is an object-typing discipline. Six quantities or roles are useful in the minimal sequence: four are directly observed, one is the latent target of the central identification question, and one is a post-episode functional category.

First, T_O is the observer-measured elapsed duration of the original bounded episode. It is defined by declared onset and offset criteria. For a self-terminated task, onset might be the end of an instruction cue and offset the participant's completion signal. T_O establishes how long that operationally bounded episode lasted; it does not establish which temporal variables were represented or used within it.

Second, U_T denotes latent online temporal use during the naïve episode. U_T asks whether metric elapsed-time information participates causally in the ongoing computation, control, comparison, prediction, stopping, or content of the target process. This is not equivalent to the physical fact that the process unfolds in time, nor is it equivalent to use of ordinal position, event sequence, rhythm, or other temporal structure that does not encode metric elapsed duration.

Third, A_{post} denotes the post-episode temporal basis, access, or reconstruction available after the original episode and after the temporal query has been introduced. It may involve retained temporal information, remembered event structure, state changes, bodily or environmental cues, priors, or active reconstruction. A_{post} is deliberately a local post-episode category, not a claim that temporal information was generally unavailable during Episode 1. Temporal information could be represented or accessible online yet not be causally recruited by the target process, and information could be causally used online without later being cleanly reportable.

The remaining directly observed quantities are the reports and the second episode. $T_{\hat{Q}}$ is the minimally elaborated post-query duration judgment about Episode 1. T_R is the observer-measured duration of Episode 2, the deliberate act of thinking about Episode 1's duration. $T_{\hat{R}}$ is the later quick duration judgment about Episode 2. The first temporal question and the reflection instruction are interventions on subsequent cognition; none of these post-query observables is a passive readout of the pre-query state.

These definitions do not equate representation, access, causal recruitment, reportability, or report accuracy. In particular, A_{post} is not the complement of U_T and Available is not defined to mean “unused.” A temporally informative state could be represented online but unused, used online but later inaccessible, accessed only after the query, reconstructed from nontemporal evidence, or combined across routes. The minimal protocol directly observes T_O , $T_{\hat{Q}}$, T_R , and $T_{\hat{R}}$; U_T remains latent, and A_{post} is a model-dependent post-episode category rather than a directly measured substance.

[Table 1 about here]

4. Why Ordinary Duration Reports Do Not Settle Online Temporal Use

A retrospective duration report is tempting to read backward. If a participant says “about 12 seconds” and the stopwatch reads 11.8 seconds, the observed pair $\{T_O, T_{\hat{Q}}\}$ is compatible with several latent structures. The participant may have monitored metric time online and then reported it; temporal information may have been represented online but not recruited by the original task and only used after the query; the participant may reconstruct duration from remembered events, bodily or environmental cues, or learned priors; or the report may combine these routes.

The converse ambiguity is equally important. If the participant reports 30 seconds for an episode that lasted 12 seconds, the discrepancy does not show that temporal information was absent online. A participant may have used metric time for one purpose while producing a biased verbal estimate later, or may have had online temporal information that was poorly accessible or mapped onto clock units. Likewise, a failure to answer is evidence of non-reportability under that probe, not evidence that temporal information was absent, unavailable, or unused.

Prospective timing reverses one ambiguity by introducing another. If the participant is told before the episode that its duration will be judged, temporal monitoring can become a goal-

relevant feature of the task. That is scientifically useful when attention to time is itself the target, but it changes the episode whose naïve temporal organization is at issue. A prospective trial therefore cannot automatically recover the same latent state that would have existed before timing became relevant.

The central theoretical payload is observational equivalence. Ordinary external duration plus a later report can be generated by multiple latent causal organizations, so those observables do not uniquely select among them. The problem is not that either measurement is defective. It is that T_O and $T_{\hat{Q}}$ answer different questions, while U_T is a latent causal property whose identification requires assumptions or evidence beyond their agreement or disagreement. Quantified-introspection and temporal-metacognition results illustrate this point empirically. Corallo et al. (2008) and Marti et al. (2010) found that subjective timing can track substantial parts of objective processing while omitting a specific objective bottleneck delay. Pavailler et al. (2025) found strong trial-level relations between objective and introspective reaction time while also showing systematic influence from nontemporal cues. Cropper et al. (2024) explicitly separated first-order duration judgment from second-order temporal metacognition. The safe inference is not that introspection is globally unreliable, but that a temporal report is evidence whose causal target must be independently established.

Perfect agreement between external and subjective duration is therefore informative but not identifying. It can show that a post-query judgment carries information about T_O . It cannot, by itself, distinguish online temporal use from online-but-unused information, post-query reconstruction, or mixed routes. To identify U_T in principle, a study would need an independently validated online manipulation or measure that is sensitive to metric temporal information during Episode 1, preserves the relevant naïveté, and makes competing latent accounts predict different outcomes. The minimal sequence below does not supply that bridge; it exposes where the bridge is missing.

5. The Naïve-Thought Protocol

The proposed protocol is designed to preserve one temporally naïve episode and then deliberately cross the boundary into temporally informed cognition. It is a future empirical test, not a report of completed participant research.

5.1. Primary sequence

The original episode should be a bounded, non-temporal thought task. “Non-temporal” means that the instruction should not ask the participant to count, estimate duration, wait for a specified interval, synchronize to a rhythm, or use time as a stopping rule. The exact content can vary across implementations, but it should be preregistered and held constant or manipulated explicitly. The participant is not told before the episode that elapsed duration is the scientific target.

The experimenter records T_O from the operational onset of the thought to the participant’s completion signal. Self-termination is useful for the simplest implementation because it allows the participant to mark when the episode feels complete, but it is not a theoretical requirement. An experimenter-terminated variant can be used when the scientific question requires fixed durations.

Only after the original episode has ended does the experimenter introduce temporal probing. The core sequence is intentionally limited to the following five steps:

1. Episode 1 occurs while the participant is temporally naïve with respect to the study target. The experimenter records its operationally defined elapsed duration as T_O .
2. After Episode 1 ends, the experimenter asks: “Give me a quick estimate of how long that took.” The response is $T_{\hat{Q}}$. “Quick” means minimally elaborated relative to the later reflection instruction; it does not mean reconstruction-free or a passive readout of a pre-existing duration representation. Response latency may be recorded as ancillary metadata.
3. The experimenter then says, approximately: “Now think about how long it took,” and starts a new timer. Duration is now explicitly task-relevant, so the participant has crossed into a temporally informed post-query regime.
4. The participant indicates when that deliberate duration-focused reflection is complete. The experimenter stops the timer and records the observer-measured duration of Episode 2 as T_R .
5. The experimenter asks for a quick estimate of how long Episode 2 itself took and records $T_{\hat{R}}$. The core sequence then stops. It intentionally does not collect a second estimate of Episode 1 after reflection.

5.2. *Preserving naïveté*

The strongest version of the first trial does not disclose before the thought that duration will be queried. Even a generic warning that timing will matter would defeat the intended contrast. The protocol’s “two questions” announcement therefore occurs only after the original thought has ended. Once the first duration question has been asked, the participant is no longer temporally naïve in the same sense. This transition is part of the design rather than a nuisance.

For this reason, repeated trials cannot all be interpreted as surprise-retrospective trials. If a within-participant study continues after the first sequence, later episodes should be labeled temporally informed or prospective with respect to duration. Researchers who need multiple naïve observations should use separate participants, counterbalanced between-subject conditions, or other designs that do not pretend that surprise can be restored after disclosure. Nicolaï et al. (2024) provide a useful precedent for treating the first retrospective trial and subsequent prospective trials differently.

5.3. *Environmental and response controls*

Eyes-closed performance is a useful optional implementation because it can reduce access to clocks, visual motion, and environmental event counts. It is not part of the theory. A lab may instead control the visual environment directly. Auditory rhythms, experimenter movements, screen animations, device vibrations, and other periodic cues should be minimized or measured if they could support either online monitoring or post-hoc reconstruction.

A participant’s body can also provide temporal cues. Heartbeat, breathing cycles, fatigue, and motor preparation may contribute to a duration estimate. The framework does not define such cues away. Their role should be manipulated or measured when relevant. The cleanest test asks whether the later report can be explained by available cues without assuming an online metric representation.

Response mechanics require similar care. If the participant presses a button to mark completion, motor preparation is part of the operational episode boundary. If the experimenter stops the timer after a verbal “done,” speech onset and device latency contribute small measurement errors. These are ordinary methodological issues, but they matter most for extremely short episodes, where boundary error can approach the duration being studied.

5.4. Thought content and episode boundaries

The protocol does not assume that “a thought” is a natural indivisible unit. It defines a bounded cognitive episode for measurement. A participant may experience several images, propositions, shifts of attention, or subgoals within that episode. Klinger (2013) and experience-sampling work make clear that natural thought boundaries are difficult to establish independently. The operational boundary should therefore be explicit and the theoretical claim limited to the bounded episode so defined.

The initial task should also avoid content that makes timing constitutive. A thought such as “count ten seconds,” “wait until your third breath,” or “decide when a minute has passed” makes U_T part of the instruction. Such tasks are valuable positive controls but not clean tests of naïve temporal use. A task with an explicit temporal stopping rule has the same problem.

5.5. Why the simple design is useful

The design does not attempt to solve all problems of time perception in one experiment. Its value is stage separation. T_O is collected without requiring a duration judgment; T-hat_Q is generated only after the first temporal query; T_R measures the duration of the deliberately induced reflection episode; and T-hat_R is generated only after Episode 2 ends. U_T remains latent, and A_post names post-query bases or operations that may contribute to reporting. The sequence therefore exposes where new cognitive operations enter without pretending that the four observables uniquely identify those latent variables.

6. Deliberate Reflection and the Recursive Probe

After T-hat_Q, the instruction “Now think about how long it took” deliberately creates Episode 2. Temporal duration is now explicit. The participant may replay events, count remembered steps, consult bodily cues, compare with familiar durations, access retained temporal information, or use some mixture of these operations. T_R measures only how long this reflection episode lasts between declared markers. It is not a measure of the amount, quality, or success of reconstruction.

The core protocol does not ask for a second estimate of Episode 1 after reflection. That omission is intentional. By then the participant has already produced T-hat_Q, knows exactly which temporal property is under examination, and has been instructed to reason about it. A later re-

estimate of Episode 1 would therefore be vulnerable to anchoring on the first response, consistency pressure, demand effects, and explicit temporal reconstruction. More importantly, the identification argument does not require such a response, so adding it would create a new interpretive problem without repairing the original one.

The recursive probe instead treats Episode 2 as the new target. The experimenter observes T_R and then obtains $\hat{T}_R$, a later estimate of Episode 2. The same caution applies one level higher: agreement between T_R and $\hat{T}_R$ does not prove that metric elapsed time was causally used during Episode 2, and disagreement does not prove that temporal information was absent or unavailable.

The role of recursion is therefore structural rather than mechanistically discriminating. Episode 2 shows that the operation introduced to reason about Episode 1's duration is itself a temporally extended cognitive episode. Objective timing of that episode and a later report about it recreate the same observer-duration/report non-identification. The recursive probe makes the inferential architecture visible; it does not uniquely select a timing mechanism and does not identify U_T for either episode.

One recursive level is enough. Extending the sequence indefinitely would rapidly become a trained timing task in which temporal monitoring is expected. The recursive probe is thus retained as a compact symmetry demonstration, not as an infinite regress, paradox, metaphysical result, or the primary novelty claim.

This creates an important asymmetry in failure conditions. The central non-identification claim is challenged only if a validated inferential bridge or additional assumptions legitimately collapse the latent alternatives. The added empirical utility of the staged protocol can fail more ordinarily: Episode 2 may add no interpretable structure beyond $\hat{T}_Q$, operational boundaries may be too noisy, or query/reactivity effects may dominate the quantities of interest. Such outcomes would narrow the value of this particular protocol without making T_O and a later report identical to U_T .

7. Observational Patterns and Failure Conditions

The protocol is most useful when its inferential limits are specified before data collection. Table 2 summarizes example observations and the strongest inference each can support without overreach.

A central principle is that report accuracy is not a binary test of online temporal use. An accurate $T\text{-}\hat{Q}$ is compatible with online temporal monitoring, online-but-unused temporal information that becomes relevant after the query, rapid reconstruction from remembered structure or other cues, and mixed routes. An inaccurate $T\text{-}\hat{Q}$ is compatible with absent $U\text{-}T$, but also with noisy post-query processing, scale-mapping error, boundary uncertainty, or later bias. The observable pair constrains models; it does not uniquely identify the latent route.

The cleanest evidence for $U\text{-}T$ would therefore require an additional independently validated online manipulation or measure. Such evidence would need to be demonstrably sensitive to metric temporal information during Episode 1, preserve the relevant naïveté rather than making duration the explicit target, and produce divergent predictions for accounts with versus without causal online temporal use. Developing such a measure is beyond the present minimal protocol. The protocol itself has a separate empirical failure mode as a constructive measurement proposal. Its added usefulness would be weak if the staged sequence produces no stable or interpretable information beyond an ordinary retrospective duration query, if $T\text{-}R$ and $T\text{-}\hat{R}$ do not support meaningful second-episode analysis, if operational boundaries are too noisy, or if task reactivity dominates the quantities being studied. Those outcomes would narrow the protocol's utility while leaving the underlying identifiability constraint intact.

The first temporal question must also be treated as an intervention rather than a neutral readout. $T\text{-}\hat{Q}$ may already involve post-query reconstruction before the participant responds, and the numerical response itself becomes part of the cognitive state entering Episode 2. Response latency can therefore be useful ancillary metadata, but the minimal protocol does not interpret short latency as proof of pre-query availability and does not add a second Episode 1 estimate whose change could be confounded by anchoring or consistency pressure.

A null report is similarly bounded. If a participant says "I have no idea," the result establishes non-reportability under that probe for that participant and trial. It does not establish that temporal information was absent during Episode 1, unavailable online, unused, or unavailable after the query. Follow-up work can separately study confidence, scale mapping, access, and report generation without treating the null response as a direct measure of $U\text{-}T$.

[Table 2 about here]

8. Edge Cases and Scope Conditions

8.1. *Thoughts that explicitly contain timing or counting*

If the thought itself is “count to ten seconds,” “estimate when 20 seconds has passed,” or “count ten breaths,” temporal information is task-relevant by construction. Such cases are not counterexamples to the framework; they are positive controls in which U_T is expected to be high. They are outside the cleanest domain for asking whether ordinary non-temporal thought uses metric elapsed time.

8.2. *Spontaneous temporal monitoring*

A participant may monitor time even when not instructed to do so. Musicians, athletes, meditators, people waiting for an expected event, or anyone unusually attentive to elapsed time may spontaneously recruit temporal cues. The protocol must allow this possibility rather than classifying it as contamination after the fact. A post-trial question about whether the participant noticed or monitored time can be informative, but it is still retrospective. Stronger studies could stratify participants by independently measured timing habits or use converging behavioral measures.

8.3. *External clocks, rhythms, and event sequences*

An available clock, periodic sound, screen animation, experimenter movement, or salient sequence of events can supply temporal information. That information may be used online or later during reconstruction. Removing such cues makes the interpretation cleaner; deliberately adding them can test cue dependence. The important point is to distinguish a report derived from environmental structure from evidence that the original thought internally represented metric elapsed duration.

8.4. *Explicit temporal stopping rules*

A temporal stopping rule makes time causally relevant. If the participant must stop “after roughly 15 seconds,” then successful task control requires some timing mechanism or proxy. Such a design can validate timing competence but cannot test naïve non-temporal use. A non-temporal self-termination rule - stop when the thought feels complete - better preserves the

intended contrast, while acknowledging that “completion” itself may depend on a sequence of cognitive events.

8.5. Extremely short and long episodes

Very short episodes magnify boundary and motor-latency errors. They may also be too brief for a participant to form a stable retrospective estimate in ordinary clock units. Very long episodes create a different problem: memory, event segmentation, changes in context, boredom, and external cues accumulate, providing rich bases for retrospective reconstruction. The most informative initial range is therefore long enough to separate thought from response mechanics but short enough to limit extensive event-based reconstruction. The exact range should be empirically calibrated rather than assumed by the theory.

8.6. Quick estimates and objective duration

Close agreement between $T_{\hat{Q}}$ and T_O shows that the first post-query judgment contains useful information about the observer-defined interval. Large disagreement shows an objective/report mismatch. Neither pattern alone identifies U_T , and “quick” should not be interpreted as pre-reconstructive. The first prompt can itself initiate memory search, cue integration, or other post-query processing before the response is produced.

8.7. Reflection after the first report

Episode 2 begins after $T_{\hat{Q}}$ has already been produced. The participant therefore enters deliberate reflection with knowledge of the first response and of the temporal target, so anchoring, consistency pressure, and demand characteristics can shape the reflection process. The core protocol treats that changed state as part of the post-query regime rather than trying to recover another uncontaminated estimate of Episode 1. This is also why no post-reflection re-estimate of Episode 1 is included in the core sequence.

8.8. Recursive behavior

The recursive estimate should be interpreted only against Episode 2. $T_{\hat{R}}$ may closely track T_R or diverge from it; either result describes the mapping between the observer-defined duration of the reflection episode and its later report. It does not provide a unique discriminator

among timing mechanisms, and it should not be compared with $T_{\hat{Q}}$ as though the two reports concerned the same episode.

8.9. Repeated questioning

Repeated duration questions train attention toward time. After the first surprise query, later trials can no longer be assumed to sample the same naïve state. This is a substantive transition: measurement changes the relevance structure of the task. Designs should exploit that transition directly, for example by comparing first-trial retrospective performance with later informed trials while separately modeling order and practice, rather than describing all trials as equivalent.

8.10. Ambiguous onset and termination

A thought can begin before an overt cue has been fully processed and can continue after a completion signal has been initiated. The observer's stopwatch therefore measures an operationally bounded episode, not the metaphysical beginning and end of a thought. The paper's claims require only that the boundary be declared and applied consistently. Studies that need finer temporal localization should add independent markers rather than silently treating the key press as exact cognition onset or offset.

8.11. Nested subepisodes

A bounded thought can contain subthoughts, imagery, inner speech, memory retrieval, decision stages, and shifts in attention. A single T_O does not imply that the episode is temporally homogeneous. This is another reason not to infer internal timing architecture from the scalar duration alone. Future work can apply the same object-typing discipline to nested stages if those stages can be independently bounded.

9. Implications for Timing Research

The first implication is terminological but consequential: studies should state whether a duration variable is an external interval, a latent online task variable, a post-episode basis/access/reconstruction category, or a report. The phrase “subject knew how long it took” often compresses representation, access, causal use, and reportability into one claim. Replacing it with the actual evidence - for example, “the participant produced an accurate retrospective estimate after the episode” - prevents a causal inference from being smuggled into a description.

The second implication concerns experimental order. Surprise retrospective timing is a scarce resource within a participant. Once duration has been queried, future trials occur in a different informational regime. The first trial is therefore not merely a noisy warm-up to discard. For some identification questions it is the only trial that samples cognition before temporal relevance has been experimentally introduced. Later trials remain useful, but they answer a different question. The third implication is that post-query processing should be staged and measured without being assumed to reveal a hidden quantity. Traditional retrospective timing theories already emphasize memory and contextual change. The deliberate-reflection interval makes a duration-focused post-query operation explicit and gives it an observer-defined duration, T_R . But T_R is not an “amount of reconstruction,” and the core protocol does not collect a second Episode 1 estimate from which to infer a reflection benefit. Its value is to mark the transition into an explicitly temporal cognitive episode and then reapply the same objective/report distinction to that episode. The fourth implication is methodological symmetry. External chronometry and introspective reports should constrain one another without being collapsed. Corallo et al. (2008) and Marti et al. (2010) demonstrate that introspective timing can track objective processing and still omit a particular delay. The lesson is not to privilege the stopwatch as the only meaningful measure, nor to privilege the report as direct access to internal timing. The useful science lies in the mapping between them.

The fifth implication is a boundary on novelty. Prospective and retrospective timing are established. Memory-based duration reconstruction and event-structure effects are established (e.g., Yue et al., 2026). Quantified introspective timing and selective blind spots are established (Corallo et al., 2008; Marti et al., 2010), and first-order temporal judgments with second-order temporal metacognition are established (Cropper et al., 2024; Pavailler et al., 2025). Experience-sampling methods already wrestle with reactivity and temporally extended thought. The residual contribution is therefore not any one component. It is the integrated identification structure applied to a temporally naïve bounded cognitive episode and the compact sequence that makes the transition from Episode 1 to post-query judgment, deliberate reflection, and Episode 2 self-timing explicit.

That boundary also limits the article’s philosophical reach. Nothing here decides whether temporal phenomenology is fundamentally metric, whether consciousness has a specious present, whether subjective duration is constructed from event change, or whether a dedicated timing

mechanism exists. Those are larger questions. The narrower methodological claim is that a later duration judgment cannot settle them unless the experiment specifies which temporal object the judgment identifies and what alternative pathways could have produced it.

10. Conclusion

How long is a thought? For an observer, a bounded cognitive episode can be assigned an elapsed duration between declared markers. For the person having the thought, that number does not settle whether metric temporal information was represented, accessible, causally used, or later reportable. A post-query report introduces another stage of cognition, and the basis of that report may have several possible histories.

The minimal architecture can be expressed as T_O , latent U_T , A_{post} , T_{hat_Q} , T_R , and T_{hat_R} . The proposed protocol preserves one temporally naïve Episode 1, measures T_O , obtains the minimally elaborated post-query judgment T_{hat_Q} , times an explicitly duration-focused Episode 2 as T_R , and then obtains T_{hat_R} . It deliberately stops there rather than asking for a post-reflection re-estimate of Episode 1. The recursive probe demonstrates that objective duration and later report remain non-identifying one level higher.

The proposal survives by keeping its claims narrow. Accurate retrospective reports do not prove online temporal use; inaccurate or absent reports do not prove online temporal absence. A_{post} does not mean that temporal information was unavailable online, and T_R does not measure the amount of reconstruction. Prospective instructions reveal timing under temporal relevance but alter the naïve target. The practical recommendation is therefore to type the observables and latent variables separately, stage post-query operations explicitly, and require an independently validated online bridge before inferring U_T .

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Data and Ethics Statement

No new participant data were generated or analyzed for this theoretical article. The proposed protocol is presented as a future empirical test; no ethics approval or preregistration is claimed for the present manuscript.

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Table 1. Measurement objects in the proposed framework

Quantity	Phase	Directly observed?	What it supports / cannot identify
T_O: observer-measured duration	Episode 1	Yes	Supports the elapsed interval between declared operational markers. By itself, it does not identify representation, access, or online temporal use.
U_T: latent online temporal use	During naïve Episode 1	No	Metric temporal information participates causally in the ongoing target process only if supported by an independent online bridge; T_O or a later report alone is insufficient.
A_post: post-episode temporal basis/access/reconstruction	After Episode 1 / after query	No, not as a single quantity	Names possible post-query bases or operations contributing to judgment. It does not imply temporal information was unavailable online, nor does it equal causal use.
T-hat_Q: first post-query report	After Episode 1	Yes	Supports that a duration judgment was produced after the query and may carry information about T_O. Accuracy or error does not identify U_T or prove reconstruction-free access.
T_R: observer-measured reflection duration	Episode 2	Yes	Supports how long the deliberate duration-focused reflection episode lasted. It does not measure the amount or success of reconstruction.
T-hat_R: report of reflection duration	After Episode 2	Yes	Supports the participant's later duration judgment about Episode 2. Agreement with T_R does not identify online temporal use during Episode 2.

Table 2. Example observations and the strongest warranted inference

Observed pattern	Supported interpretation	What it does not establish
T-hat_Q closely tracks T_O	The first post-query judgment carries useful information about the observer-defined duration of Episode 1.	That metric elapsed time was causally used online, or that the report was reconstruction-free.
T-hat_Q diverges from T_O	Objective and reported duration dissociate under these conditions.	That temporal information was absent, unavailable, or unused during Episode 1.
T_R varies across conditions or participants	The duration of the explicitly temporal reflection episode changes.	The amount, quality, or success of reconstruction, or improvement of an Episode 1 estimate.
T-hat_R closely tracks T_R	The post-query report about Episode 2 carries information about its observer-defined duration.	That metric elapsed time was causally used online during Episode 2.

Observed pattern	Supported interpretation	What it does not establish
T-hat_R diverges from T_R	The objective/report dissociation recurs for the reflection episode.	A paradox, absence of temporal information, or a unique timing mechanism.
The staged sequence adds no stable structure beyond T-hat_Q	The proposed protocol has weak added empirical utility in that implementation.	That T_O or T-hat_Q identifies U_T, or that the central non-identification constraint fails.
Participant cannot estimate duration	Duration is not reportable under that probe for that participant/trial.	That temporal information was absent during or after the episode.